

Research Highlights

Computers & Fluids Submission

Title: High-Resolution Finite-Difference Benchmarks for the 2D Lid-Driven Cavity Flow up to $Re = 100,000$: Exact Discrete Sine Transform, Batchelor Core Slices, and Hopf Bifurcation Cascades

Highlights (Each item strictly ≤ 125 characters including spaces):

- Zero-dissipation FDM pairing pure central differencing with an exact $\mathcal{O}(N^2 \log N)$ DST-I Poisson engine.
- Steady continuation up to $Re = 100,000$ on 1025^2 mesh verifying Batchelor's core theorem ($\omega_0 = -2.5418$).
- Spalding–Patankar hybrid differencing arrests corner shocks ($Pe \gg 2$) while preserving $\nu_{\text{num}} = 0$ in the core.
- Time-accurate marching maps supercritical Hopf, 2-torus, and limit-cycle regimes up to $Re = 100,000$ ($St = 0.5128$).
- Open-source release of 28 high- Re flow field tensors, limit-cycle orbits, and automated visualization pipelines.